

Propoziție 1. Fie $A \in \mathcal{J}(\mathbb{R}^n)$ cu $\lambda(A) = 0$. Atunci orice funcție $f: A \rightarrow \mathbb{R}$ mărginită este integrabilă Riemann pe A și
$$\int_A f(x) dx = 0.$$

Demonstratie (exercițiu).

Propoziție 2 Fie $A, B \in \mathcal{J}(\mathbb{R}^n)$ și $f: A \cup B \rightarrow \mathbb{R}$ mărginită și integrabilă Riemann pe A și pe B . Atunci f este integrabilă Riemann pe $A \cup B$ și pe $A \cap B$ și

$$\int_{A \cup B} f(x) dx = \int_A f(x) dx + \int_B f(x) dx - \int_{A \cap B} f(x) dx.$$

Corolar 1. Fie $A, B \in \mathcal{J}(\mathbb{R}^n)$ a.i. $\lambda(A \cap B) = 0$ si $f: A \cup B \rightarrow \mathbb{R}$

mărginită si integrabilă Riemann pe A si pe B . Atunci f este integrabilă Riemann pe $A \cup B$ si

$$\int_{A \cup B} f(x) dx = \int_A f(x) dx + \int_B f(x) dx$$

Propozitie 3 Fie $A, B \in \mathcal{J}(\mathbb{R}^n)$ a.i. $\lambda(B) = 0$ si

$f: A \cup B \rightarrow \mathbb{R}$ mărginită. Atunci f este integrabilă

Riemann pe $A \cup B$ dacă si numai dacă f este

int. Riemann pe A . In acest caz $\int_{A \cup B} f(x) dx = \int_A f(x) dx.$

Schimbarea de variabilă pt integrală multiplă

Teoremă Fie $U, V \subset \mathbb{R}^n$ mulțimi deschise,
 $\phi: U \rightarrow V$ un difeomorfism de clasă C^1 (schimbare de
variabilă) și $A \subset V$, $A \in \mathcal{J}(\mathbb{R}^n)$. Atunci

1) $\phi^{-1}(A) \in \mathcal{J}(\mathbb{R}^n)$

2) Pt. orice funcție $f: A \rightarrow \mathbb{R}$ integrabilă Riemann
funcția $f \circ \phi: \phi^{-1}(A) \rightarrow \mathbb{R}$ este integrabilă Riemann

$$\int_A f(x) dx = \int_{\phi^{-1}(A)} f \circ \phi(u) \cdot |\det(J_\phi(u))| du.$$

unde $\det(J_\phi(u))$ este Jacobianul lui ϕ în $u \in U$.

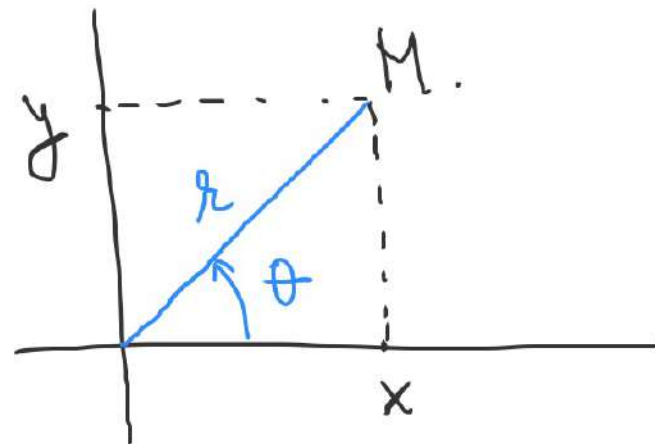
Obs.

$$\int_a^b f \circ \phi(u) \cdot \phi'(u) du = \int_{\phi(a)}^{\phi(b)} f(x) dx, \quad dx = \phi'(u) du$$

$$[a, b] \xrightarrow{\phi} [\phi(a), \phi(b)] \longrightarrow \mathbb{R}.$$

1) Treceam la coordonate polare.

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad x^2 + y^2 = r^2, \quad r \geq 0.$$



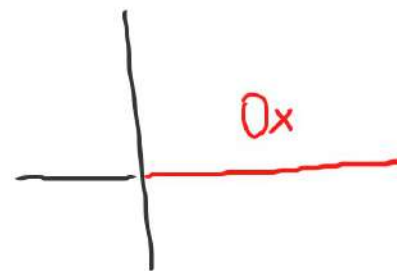
$$\phi: [0, \infty) \times [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$\phi(r, \theta) = (r \cos \theta, r \sin \theta)$$

$$\phi: \begin{cases} x = r \cos \theta & x(r, \theta) \\ y = r \sin \theta & y(r, \theta) \end{cases}$$

ϕ - surjectivă.

Restricția lui ϕ la $(0, \infty) \times (0, 2\pi)$ este un difeomorfism de clasă C^1 între $(0, \infty) \times (0, 2\pi)$ și $\mathbb{R}^2 \setminus \partial X$ unde $\partial X = \{(x, 0) \mid x \geq 0\}$



$(0, \infty) \times (0, 2\pi) \xleftarrow{\phi} \mathbb{R}^2 \setminus O_X = \mathbb{R}^2 \setminus \{(x, 0) \mid x \geq 0\}$ difeom de
 clasă C^1 .

$$\det J_{\phi}(r, \theta) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

$\phi; \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$

$$\det(J_{\phi}(r, \theta)) = \frac{D(x, y)}{J(r, \theta)} = r \neq 0, \quad \forall (r, \theta) \in (0, \infty) \times (0, 2\pi)$$

$A \in \mathcal{J}(\mathbb{R}^2)$, $f: A \rightarrow \mathbb{R}$ integrabilità Riemann

$$A' = A \setminus O_x$$

$$\lambda(A') = \lambda(A) , \lambda(A \setminus A') = 0.$$

$$\lambda(\phi^{-1}(A) \setminus \phi^{-1}(A')) = \lambda(\phi^{-1}(A \cap O_x)) = 0.$$

$$\left| \frac{D(x,y)}{D(r,\theta)} \right| = r.$$

$$\iint_{A'} f(x,y) dx dy = \iint_{\phi^{-1}(A')} f(r \cos \theta, r \sin \theta) \cdot \underbrace{|\det J_\phi(r,\theta)|}_{r} dr d\theta$$

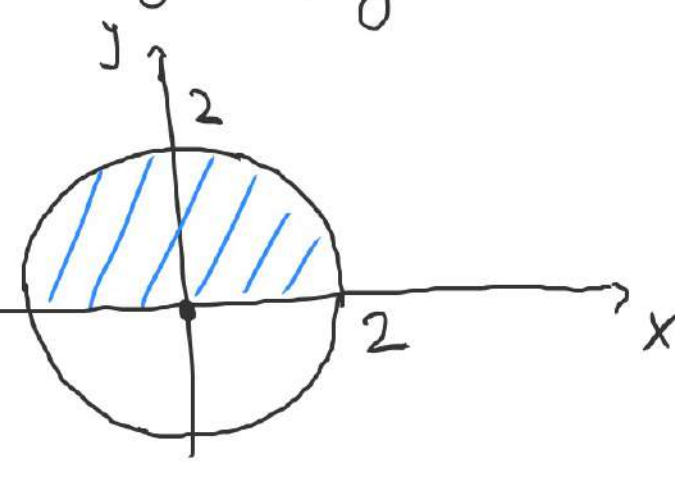
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$$\iint_A f(x,y) dx dy = \iint_{\phi^{-1}(A)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta.$$

Calculati $\iint_D x^2 dx dy$, $D = \{(x, y) \mid x^2 + y^2 \leq 4, y \geq 0\}$

$D = \{(x, y) \mid -2 \leq x \leq 2, 0 \leq y \leq \sqrt{4-x^2}\} \in \mathcal{J}(\mathbb{R}^2)$

cont



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$(x, y) \in D \iff \begin{cases} r \in [0, 2] \\ \theta \in [0, \pi] \end{cases}$$

$$\begin{aligned} \phi(r, \theta) &= (r \cos \theta, r \sin \theta) \\ \phi^{-1}(D) &= [0, 2] \times [0, \pi] \end{aligned}$$

$$\left(\begin{aligned} x^2 + y^2 = r^2 \leq 4 &\iff r \in [0, 2] \\ r \sin \theta \geq 0 &\iff \theta \in [0, \pi] \end{aligned} \right)$$

$$|\det J| = \left| \frac{D(x, y)}{D(r, \theta)} \right| = r \quad \left(dx dy = |\det J| dr d\theta = r dr d\theta \right)$$

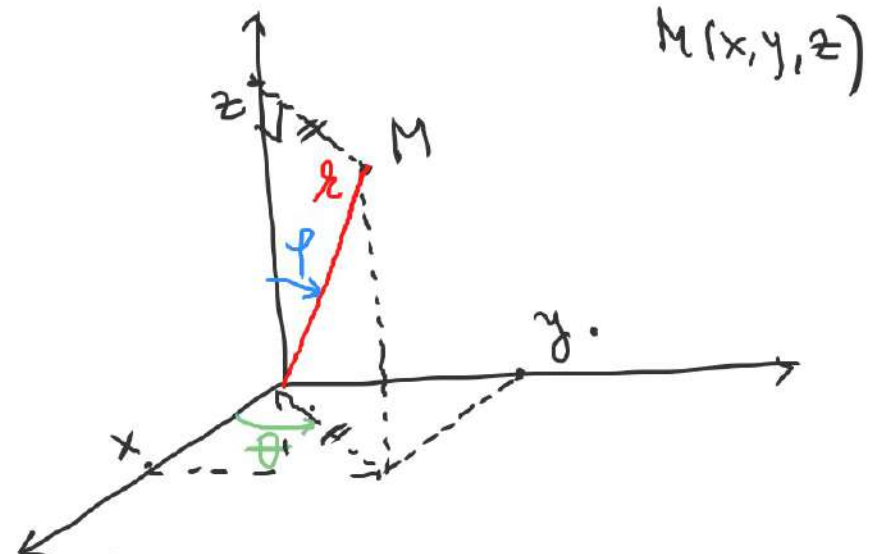
$$\iint_D x^2 dx dy = \iint_{[0,2] \times [0,\pi]} r^2 \cos^2 \theta \cdot r dr d\theta = \int_0^2 \left(\int_0^\pi r^3 \cos^2 \theta d\theta \right) dr =$$

$$\cos 2\theta = 2\cos^2 \theta - 1 \Rightarrow \cos^2 \theta = \frac{\cos 2\theta + 1}{2}$$

$$= \int_0^2 \left(\int_0^\pi r^3 \cdot \frac{\cos 2\theta + 1}{2} d\theta \right) dr = \int_0^2 r^3 \left(\frac{\sin 2\theta}{4} + \frac{\theta}{2} \right) \Big|_0^\pi dr = \int_0^2 \frac{\pi r^3}{2} dr = 2\pi$$

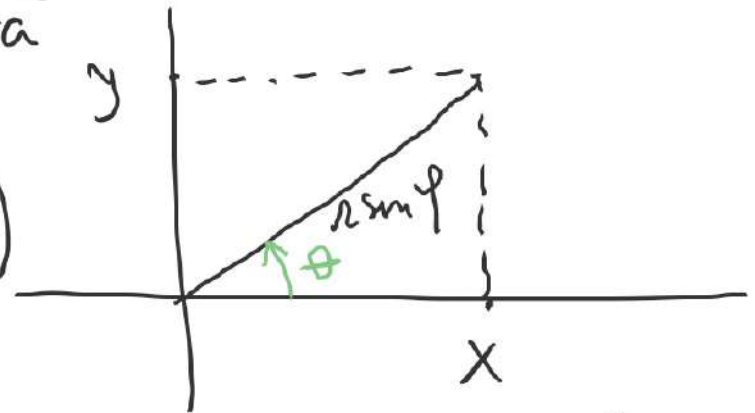
2) Trece la coordonate sferice.

$$\begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases} \quad x^2 + y^2 + z^2 = r^2$$



$\phi: [0, \infty) \times [0, 2\pi] \times [0, \pi] \rightarrow \mathbb{R}^3$ surjectivă

$$\phi(r, \theta, \varphi) = (r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi)$$



$$(0, \infty) \times (0, 2\pi) \times (0, \pi) \xleftarrow{\phi} \mathbb{R}^3 \setminus \{(x, 0, z) \mid x \geq 0, z \in \mathbb{R}\}$$

difeomorfism de
clasă C^1 .

$$S = \{(x, 0, z) \mid x \geq 0, z \in \mathbb{R}\}$$

$$\det J_{\phi}(r, \theta, \varphi) = \frac{D(x, y, z)}{D(r, \theta, \varphi)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \varphi} \end{vmatrix}$$

$$= \begin{vmatrix} \cos \theta \sin \varphi & -r \sin \theta \sin \varphi & r \cos \theta \cos \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \varphi & 0 & -r \sin \varphi \end{vmatrix} =$$

$$= - \underline{r^2 \cos^2 \theta \sin^3 \varphi} - r^2 \sin^2 \theta \cos^2 \varphi \sin \varphi - r^2 \cos^2 \theta \cos^2 \varphi \sin \varphi -$$

$$- \underline{r^2 \sin^2 \theta \sin^3 \varphi} = -r^2 \sin^3 \varphi - r^2 \cos^2 \varphi \sin \varphi = -r^2 \sin \varphi.$$

$$|\det J_\phi(r, \theta, \varphi)| = \left| \frac{D(x, y, z)}{D(r, \theta, \varphi)} \right| = r^2 \sin \varphi.$$

$$dx dy dz = r^2 \sin \varphi dr d\theta d\varphi$$

$V \in \mathcal{I}(\mathbb{R}^3)$, $f: V \rightarrow \mathbb{R}$ integrable Riemann

$$V' = V \setminus S, \quad \lambda(V \setminus V') = 0$$

$$\lambda(\phi^{-1}(V) \setminus \phi^{-1}(V')) = 0.$$

$$\underbrace{\iiint_{V'} f(x, y, z) dx dy dz}_{\parallel} = \iiint_{\phi^{-1}(V')} f(r \cos \theta \sin \varphi, r \sin \theta \sin \varphi, r \cos \varphi) \overbrace{r^2 \sin \varphi dr d\theta d\varphi}^{dx dy dz}$$

$$\iiint_V f(x, y, z) dx dy dz = \iiint_{\phi^{-1}(V)} f(r \cos \theta \sin \varphi, r \sin \theta \sin \varphi, r \cos \varphi) r^2 \sin \varphi dr d\theta d\varphi$$

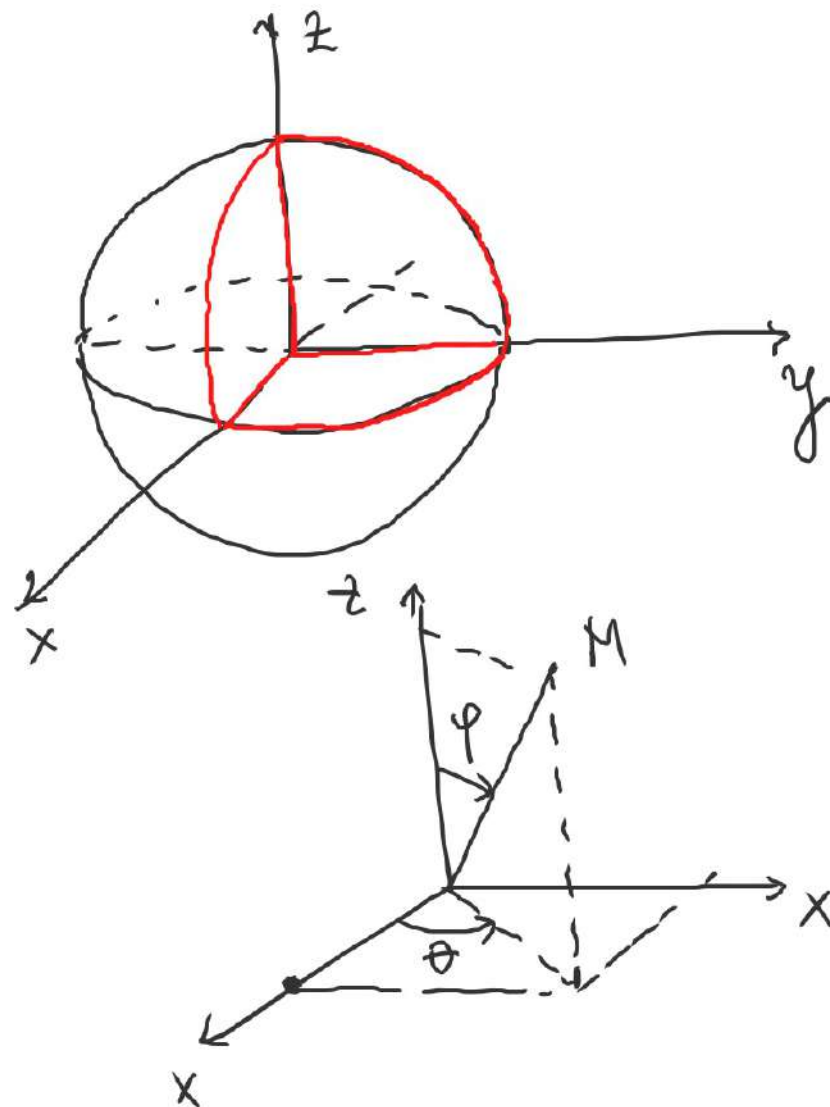
$$\iiint_V x^2 dx dy dz \quad V = \{ (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 4, x, y, z \geq 0 \} \in \mathcal{J}(\mathbb{R}^3) (\text{ex.})$$

$$\begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases}$$

$$(x, y, z) \in V \Leftrightarrow \begin{cases} r \in [0, 2] \\ \theta \in [0, \frac{\pi}{2}] \\ \varphi \in [0, \frac{\pi}{2}] \end{cases}$$

$$|\det J| = r^2 \sin \varphi$$

$$dx dy dz = r^2 \sin \varphi dr d\theta d\varphi$$



$$\iiint_V x^2 dx dy dz = \iiint_{[0,2] \times [0,\frac{\pi}{2}] \times [0,\frac{\pi}{2}]} r^2 \sin^2 \varphi \cos^2 \theta \cdot r^2 \sin \varphi dr d\theta d\varphi$$

$$= \int_0^2 \left(\int_0^{\frac{\pi}{2}} \left(\int_0^{\frac{\pi}{2}} r^4 \sin^3 \varphi \cos^2 \theta d\varphi \right) d\theta \right) dr$$

$$= \left(\int_0^2 r^4 dr \right) \cdot \left(\int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \right) \cdot \left(\int_0^{\frac{\pi}{2}} \sin^3 \varphi d\varphi \right) = \dots$$

Exercitii

1*) Fie $A \in \mathcal{J}(\mathbb{R}^n)$ cu $\lambda(A) = 0$ si $f: A \rightarrow \mathbb{R}$ marginita.

Aratati ca f este integrabila Riemann, si $\int_A f dx = 0$.

2*) Fie $f: [a, b] \times [a, b] \rightarrow \mathbb{R}$ continua. Aratati ca

$$\int_a^b \left(\int_a^y f(x, y) dx \right) dy = \int_a^b \left(\int_x^b f(x, y) dy \right) dx$$

3) Calculati $\iint_D \sqrt{x^2 + y^2} dx dy$; $D = \{(x, y) \in \mathbb{R}^2 \mid 2x \leq x^2 + y^2 \leq 4x, y \geq 0\}$

$\iiint_V y dx dy dz$; $V = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 4y - z \leq 8, x, y \geq 0, z \leq -4\}$